

QCT Supplementary Material 13

Dark Matter

Content Pending, Publication-Ready Shell

Andrew Lizzio

May 7, 2026

Abstract

Quantum Cell Theory (QCT) is a foundational framework derived from three postulates: **P1** (the universe comprises a finite number N of discrete energy cells, each carrying rest energy ε_0 , with cell count and vibrational energy conserved); **P2** (each cell resizes in proportion to local vibrational energy density, isotropically and scale-free); and **P3** (one time quantum τ_0 elapses per wave-propagation step).

QCT predicts dark matter candidates from higher-charge soliton sectors ($Q > 1$), with mass predictions following from the same D_5 geometry that determines the electron and proton masses. This document will contain QCT derivations of dark matter candidates, their mass predictions, and the connection to the cosmological dark matter fraction derived in QCT-Supp-5-Cosmology-v1. Derivations are in progress; this shell establishes the module structure.

Contents

1	Higher-Charge Solitons as Dark Matter	4
1.1	$Q > 1$ Hopf Solitons in QCT	4
1.2	Stability of Higher-Charge Sectors	4
2	Dark Matter Mass Predictions	4
2.1	$Q = 2$ Soliton Mass from D_5 Geometry	4
2.2	Mass Hierarchy Across Soliton Sectors	4
3	The Cosmological Dark Matter Fraction	4
3.1	Connection to $\Omega_{\text{CDM}}/\Omega_b = 5.41$ from Supp-5	4
3.2	QCT Dark Matter Density Profile	4
4	Observational Signatures	4
4.1	Direct Detection Cross-Sections from QCT	4
4.2	Indirect Detection Signatures	4

About This Module and the QCT Series

This document is part of the QCT Supplementary series of standalone modules. The series is:

- **QCT-Supp-0-Map-v1:** Proof Map
- **QCT-Supp-1-Postulates-v1:** Postulates P1/P2/P3, Classical Mechanics, General Relativity, Quantum Mechanics
- **QCT-Supp-2-Identification-v1:** Structural Identifications ID1/ID2/ID3, Fine Structure Constant α
- **QCT-Supp-3-MassGap-v1:** Yang–Mills Existence and Mass Gap
- **QCT-Supp-4-ElectronMass-v1:** Hopf Solitons, $N_c = 20$, Electron Mass Formula
- **QCT-Supp-5-Cosmology-v1:** Cosmological derivations C1–C11, Dark Matter and Dark Energy
- **QCT-Supp-6-Spectral-v1:** Electromagnetism, Force Mechanism, Weinberg Angle
- **QCT-Supp-7-Chemistry-v1:** Chemical and Molecular Predictions
- **QCT-Supp-8-ParticleMasses-v1:** Particle Mass Spectrum
- **QCT-Supp-9-NuclearPhysics-v1:** Nuclear Physics
- **QCT-Supp-10-Neutrinos-v1:** Neutrino Physics
- **QCT-Supp-11-QuantumGravity-v1:** Quantum Gravity and Planck Scale
- **QCT-Supp-12-DarkMatter-v1:** Dark Matter

- 1 Higher-Charge Solitons as Dark Matter**
 - 1.1 $Q > 1$ Hopf Solitons in QCT**
 - 1.2 Stability of Higher-Charge Sectors**
- 2 Dark Matter Mass Predictions**
 - 2.1 $Q = 2$ Soliton Mass from D_5 Geometry**
 - 2.2 Mass Hierarchy Across Soliton Sectors**
- 3 The Cosmological Dark Matter Fraction**
 - 3.1 Connection to $\Omega_{\text{CDM}}/\Omega_b = 5.41$ from Supp-5**
 - 3.2 QCT Dark Matter Density Profile**
- 4 Observational Signatures**
 - 4.1 Direct Detection Cross-Sections from QCT**
 - 4.2 Indirect Detection Signatures**